

Mithesh Kosanam

SPL-MS

DBMS

Smart Parking Lot Management System

Woxsen University

PROBLEM STATEMENT

Time-Consuming Operations

Manual lot management introduces delays and human error at every stage of the parking workflow.

No Real-Time Visibility

Staff and drivers have no live insight into which spaces are available, causing congestion and frustration.

Manual Billing & Reservations

Tracking reservations and calculating charges by hand leads to billing errors and revenue loss.

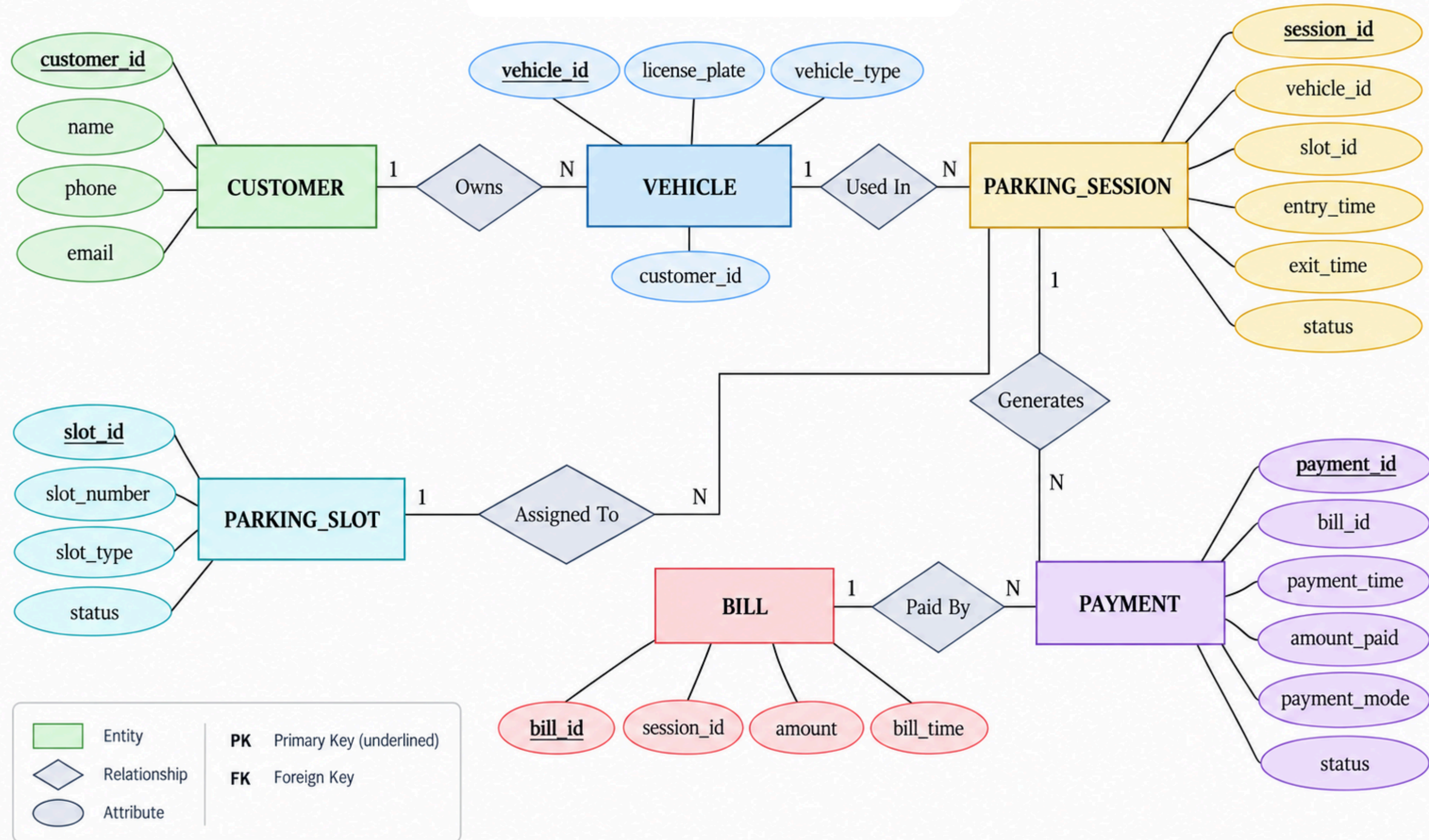
No Analytics Capability

Without structured data, it is impossible to analyze usage patterns or make informed operational decisions.

OBJECTIVE

- Develop a centralized database system for efficient smart parking lot management.
- Manage parking facilities, floors, zones, slots, vehicles, customers, and vehicle types.
- Manage parking passes, bills, payments, and reservation expiry.
- Record vehicle entry and exit accurately to prevent slot conflicts and unauthorized usage.
- Automate parking fee calculation based on parking duration and applicable tariffs.
- Enable real-time slot search, allocation, reservation, and availability tracking.

SMART PARKING LOT



DATA BASE DESIGN

Table	Primary Key	Foreign Keys
CUSTOMER	customer_id	—
VEHICLE	vehicle_id	customer_id
PARKING_SLOT	slot_id	—
PARKING_SESSION	session_id	vehicle_id, slot_id
BILL	bill_id	session_id
PAYMENT	payment_id	bill_id

END

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